LINKED LIST

Q.1 Write a program in C to create and display Singly Linked List.

Test Data:

Input the number of nodes: 3
Input data for node 1: 5
Input data for node 2: 6
Input data for node 3: 7

Expected Output:

Data entered in the list:
Data = 5
Data = 6
Data = 7

Q.2 Write a program in C to create a singly linked list of n nodes and display it in reverse order.

Test Data:

Input the number of nodes : 3
Input data for node 1 : 5
Input data for node 2 : 6
Input data for node 3 : 7

Expected Output :

Data entered in the list are :
Data = 5
Data = 6
Data = 7
The list in reverse are :
Data = 7
Data = 6
Data = 5

Q.3. Write a program in C to create a singly linked list of n nodes and count the number of nodes.

Test Data :

Input the number of nodes : 3
Input data for node 1 : 5
Input data for node 2 : 6
Input data for node 3 : 7

Expected Output :

Data entered in the list are :
Data = 5
Data = 6
Data = 7
Total number of nodes = 3

Q.4 Write a program in C to insert a new node at the beginning of a Singly Linked List.

Test Data and Expected Output **:**

Input the number of nodes : 3
Input data for node 1 : 5
Input data for node 2 : 6
Input data for node 3 : 7

Data entered in the list are :
Data = 5
Data = 6
Data = 7
Input data to insert at the beginning of the list : 4

Data after inserted in the list are :
Data = 4
Data = 5
Data = 6
Data = 7

Q.5. Write a program in C to insert a new node at the end of a Singly Linked List.

Test Data and Expected Output :

Input the number of nodes : 3
Input data for node 1 : 5
Input data for node 2 : 6
Input data for node 3 : 7

Data entered in the list are :
Data = 5
Data = 6
Data = 7

Input data to insert at the end of the list : 8

Data, after inserted in the list are :
Data = 5
Data = 6
Data = 7
Data = 8

Q.6. Write a program in C to insert a new node at the middle of Singly Linked List.

***Test Data and Expected Output* :**

Input the number of nodes (3 or more) : 4
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3
Input data for node 4 : 4

Data entered in the list are :
Data = 1
Data = 2
Data = 3
Data = 4

Input data to insert in the middle of the list : 5
Input the position to insert new node : 3

Insertion completed successfully.

The new list are :
Data = 1
Data = 2
Data = 5
Data = 3
Data = 4

Q.7. Write a program in C to delete first node of Singly Linked List.

Test Data :

Input the number of nodes : 3

Input data for node 1 : 2

Input data for node 2 : 3

Input data for node 3 : 4

Expected Output :

Data entered in the list are :
Data = 2
Data = 3
Data = 4

Data of node 1 which is being deleted is : 2

Data, after deletion of first node :
Data = 3
Data = 4

Q.8. Write a program in C to delete a node from the middle of Singly Linked List.

***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data = 2
Data = 5
Data = 8

Input the position of node to delete : 2

Deletion completed successfully.

The new list are :
Data = 2
Data = 8

Q.9. Write a program in C to delete the last node of Singly Linked List.

Test Data :

Input the number of nodes : 3

Input data for node 1 : 1

Input data for node 2 : 2

Input data for node 3 : 3

Expected Output :

Data entered in the list are :
Data = 1
Data = 2
Data = 3
The new list after deletion the last node are :
Data = 1
Data = 2

Q.10. Write a program in C to search an existing element in a singly linked list.

Test Data and Expected

Input the number of nodes : 3

Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data = 2
Data = 5
Data = 8

Input the element to be searched : 5
Element found at node 2

Q.11. Write a program in C to create and display a doubly linked list.

Test Data :

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Expected Output :

Data entered on the list are :
node 1 : 2
node 2 : 5
node 3 : 8

Q.12 Write a program in C to create a doubly linked list and display in reverse order.

Test Data :

Input the number of nodes : 3

Input data for node 1 : 2

Input data for node 2 : 5

Input data for node 3 : 8

Expected Output :

Data in reverse order are :
Data in node 1 : 8
Data in node 2 : 5
Data in node 3 : 2

Q.13 Write a program in C to insert a new node at the beginning in a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
node 1 : 2
node 2 : 5
node 3 : 8
Input data for the first node : 1

After insertion the new list are :
node 1 : 1
node 2 : 2
node 3 : 5
node 4 : 8

Q.14 Write a program in C to insert a new node at the end of a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
node 1 : 2
node 2 : 5
node 3 : 8
Input data for the last node : 9

After insertion the new list are :
node 1 : 2
node 2 : 5
node 3 : 8
node 4 : 9

Q.15. Write a program in C to insert a new node at any position in a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 2
Input data for node 2 : 4
Input data for node 3 : 5

Data entered in the list are :
node 1 : 2
node 2 : 4
node 3 : 5
Input the position (2 to 2) to insert a new node : 2
Input data for the position 2 : 3

After insertion the new list are :
node 1 : 2
node 2 : 3
node 3 : 4
node 4 : 5

16. Write a program in C to insert a new node at the middle in a doubly linked list.

***Test Data and Expected Output* :**

Doubly Linked List :
Insert new node at the middle in a doubly linked list :
Input the number of nodes (3 or more): 3
Input data for node 1 : 2
Input data for node 2 : 4
Input data for node 3 : 5

Data entered in the list are :
node 1 : 2
node 2 : 4
node 3 : 5
Input the position (2 to 2) to insert a new node : 2
Input data for the position 2 : 3

After insertion the new list are :
node 1 : 2
node 2 : 3
node 3 : 4
node 4 : 5

Q.17. Write a program in C to delete a node from the beginning of a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3

After deletion the new list are :
node 1 : 2
node 2 : 3

18. Write a program in C to delete a node from the last of a doubly linked list.

***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3

After deletion the new list are :
node 1 : 1
node 2 : 2

Q.19. Write a program in C to delete a node from any position of a doubly linked list.
***Test Data and Expected Output* :**

Doubly Linked List :

Delete node from any position of a doubly linked list :

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3
Input the position (1 to 3) to delete a node : 3

After deletion the new list are :
node 1 : 1
node 2 : 2

Q.20. Write a program in C to delete a node from the middle of a doubly linked list.
***Test Data and Expected Output* :**

Input the number of nodes (3 or more): 3
Input data for node 1 : 1
Input data for node 2 : 2
Input data for node 3 : 3

Data entered in the list are :
node 1 : 1
node 2 : 2
node 3 : 3
Input the position (1 to 3) to delete a node : 2

After deletion the new list are :
node 1 : 1
node 2 : 3

Q.21. Write a program in C to find the maximum value from a doubly linked list.
Test Data :

Input the number of nodes : 3

Input data for node 1 : 5

Input data for node 2 : 9

Input data for node 3 : 1

Expected Output :

Data entered in the list are :
node 1 : 5
node 2 : 9
node 3 : 1

The Maximum Value in the Linked List : 9

Q.22. Write a program in C to create and display a circular linked list.

Test Data :

Input the number of nodes : 3

Input data for node 1 : 2

Input data for node 2 : 5

Input data for node 3 : 8

Expected Output :

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

Q.23. Write a program in C to insert a node at the beginning of a circular linked list.

Test Data and Expected Output **:**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Input data to be inserted at the beginning : 1

After insertion the new list are :
Data 1 = 1
Data 2 = 2
Data 3 = 5
Data 4 = 8

Q.24. Write a program in C to insert a node at the end of a circular linked list.

Test Data and Expected Output **:**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Input the data to be inserted : 9

After insertion the new list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Data 4 = 9

Q.25. Write a program in C to insert a node at any position in a circular linked list.

***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8
Input the position to insert a new node : 3
Input data for the position 3 : 7

After insertion the new list are :
Data 1 = 2
Data 2 = 5
Data 3 = 7
Data 4 = 8

Q.26. Write a program in C to delete node from the beginning of a circular linked list.

Test Data :

Input the number of nodes : 3

Input data for node 1 : 2

Input data for node 2 : 5

Input data for node 3 : 8

Expected Output :

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

The deleted node is -> 2
After deletion the new list are :
Data 1 = 5
Data 2 = 8

Q.27. Write a program in C to delete a node from the middle of a circular linked list.

***Test Data and Expected Output* :**

Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

Input the position to delete the node : 3

The deleted node is : 8
After deletion the new list are :
Data 1 = 2
Data 2 = 5

Q.28. Write a program in C to delete the node at the end of a circular linked list.

Test Data and Expected Output :
Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 8

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 8

The deleted node is : 8
After deletion the new list are :
Data 1 = 2
Data 2 = 5

Q.29. Write a program in C to search an element in a circular linked list.

***Test Data and Expected Output* :**

Circular Linked List : Search an element in a circular linked list :
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Input the number of nodes : 3
Input data for node 1 : 2
Input data for node 2 : 5
Input data for node 3 : 9

Data entered in the list are :
Data 1 = 2
Data 2 = 5
Data 3 = 9
Input the element you want to find : 5
Element found at node 2

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